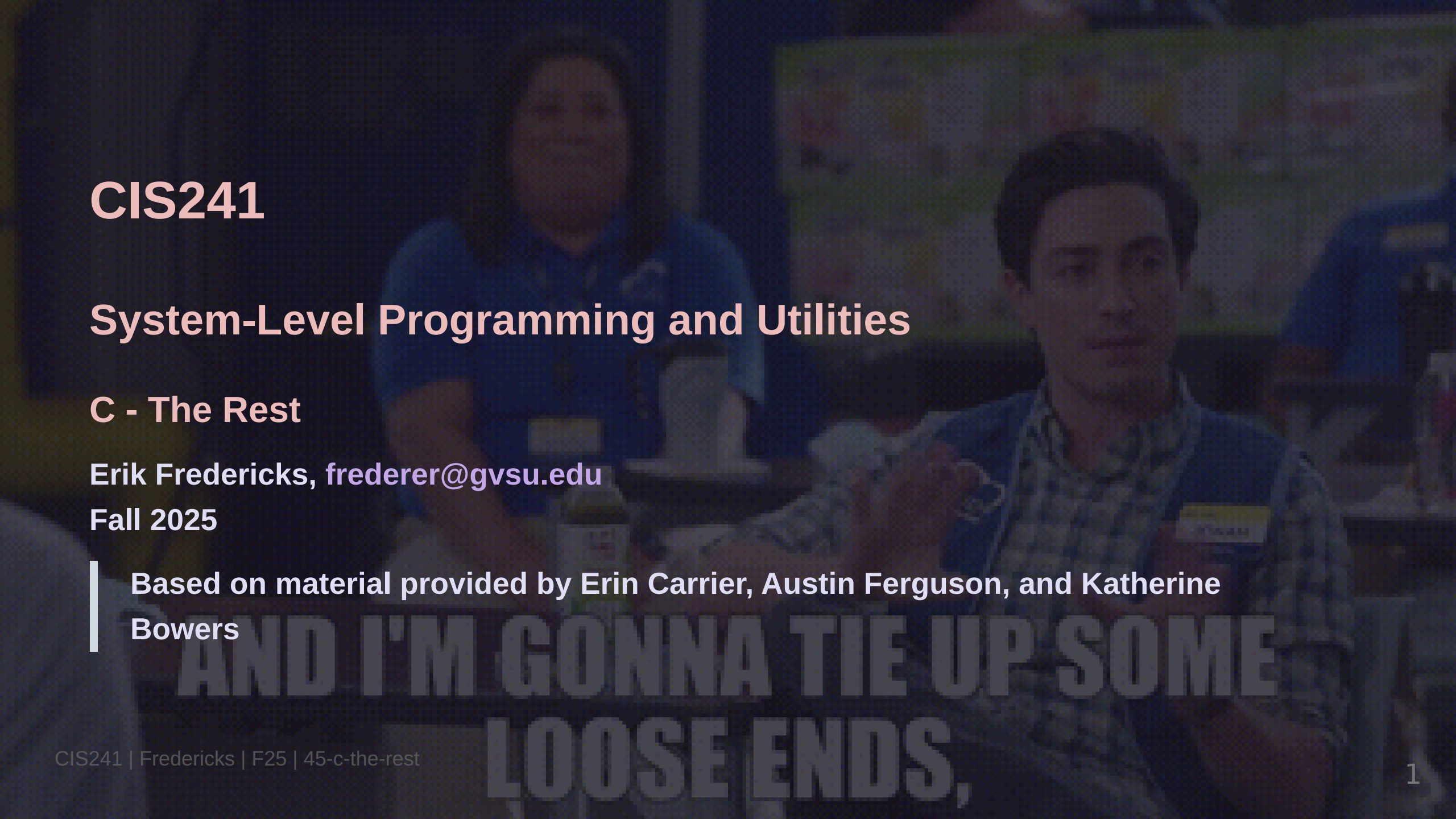


The background of the slide is a dark, low-key photograph of two people, a woman and a man, sitting at a table. They appear to be in a workshop or classroom setting. A large, semi-transparent meme text is overlaid on the bottom half of the image.

CIS241

System-Level Programming and Utilities

C - The Rest

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

Enumerations (enums)

Enums let you name "options"

- Can increase readability
- ... but doesn't actually add any new functionality

All values are just an integer underneath

- Typically start at 0 and count up
- But we can override that if we want!

```
enum rank {  
    CAPTAIN,  
    FIRST_MATE,  
    QUARTERMASTER  
};  
enum rank my_rank = QUARTERMASTER;
```

typedef (again)

What does typedef do?

- Gives an existing type a new name!
- We can use that here to trim down 'enum rank'

```
typedef enum rank {  
    CAPTAIN = 5,  
    FIRST_MATE = 4,  
    QUARTERMASTER = 3  
} rank;  
rank your_rank = CAPTAIN;
```

Unions

What does a struct do?

- Store multiple pieces of data together as a bundle

A union does the opposite: Stores multiple types of data in the same place

- All types use the same memory
- Can only store one at a time
- You are responsible for interpreting it!

Unions

```
union Data {  
    int i;  
    double d;  
    char c;  
};  
union Data var;  
var.d = 3.14;
```

Other keywords

A variable preceded by `const` cannot be changed

- Can make code more readable
- Compiler keeps you from changing it
- Optimizations
- Be careful with "const pointers"
 - <https://stackoverflow.com/questions/21476869/constant-pointer-vs-pointer-to-constant>

A variable preceded by `static` maintains its value outside of the normal scope

- E.g., a `static int` in a function will have the same value across function calls

Other libraries

Working with standard libraries is easy

- No extra work
- E.g., `stdio.h`

You can also use custom libraries

- May need to pass additional flags to gcc
- Linker flags: `-l` or `-L` options
- Includes: `-I` options

And so

